

Austin Jiang

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Dear hiring manager,

I am a Computer Science student at the University of Waterloo applying for the Software Engineering Intern role. My interests lie in performance engineering, concurrent systems, and machine learning systems.

I am currently an undergraduate research assistant in the Multicore Lab at the University of Waterloo, where I work on benchmarking and extending the **Verlib (PPoPP'24)** system. My work includes migrating the build to the CUDA toolchain to resolve host-device compatibility issues, developing an adapter supporting versioned and non-versioned data structures, and benchmarking range queries at **10M-key scale under 100+ threads**. I also implemented B+ Tree and ART data structures using the GPU Range Query Dictionary library for performance comparison.

Previously at Wolfram Research, I developed a multi-threaded engine managing over 800K data points across 8 CPU cores and designed sparse frontier updates and dense scans, achieving a **127% speedup** through improved load balancing. I also built a benchmarking harness measuring throughput. My early work has also produced a first-author research paper presented at the Wolfram Technology Conference 2025.

Recently, I was shortlisted for **Mistral AI's core research internship**. Although the process was paused due to headcount, the technical evaluation confirmed my understanding of KV cache optimization, RoPE implementations, and memory-efficient Transformers. In addition, my TreeHacks project involved deploying Hunyuan WorldPlay 1.5 across 4× H200 GPUs using tensor parallelism.

My competitive programming background, including being **two-time Canadian Computing Olympiad top 10 nationally** and USA Computing Olympiad highest division, showcases my strong problem-solving ability in algorithms and data structures.

I would love the opportunity to contribute to your team.

Sincerely,
Austin Jiang